

Audrey Reinhard

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EDUCATION

B.S + M.S. in Computer Engineering GPA - 3.72/4.00 University of California, Riverside June 2027

Honors: UC Regents Scholar, NSF S-STEM Scholarship

Relevant Coursework: Embedded Systems, Electronic Circuits, Logic Design, Computer Architecture, GPU Programming

SKILLS

Embedded Systems: C, C++, Arduino, ESP32, Raspberry Pi, ATmega, UART, SPI, I2C, CAN, PWM, Sensors, Actuators

Robotics & AI: ROS2, OpenCV, TensorFlow, BERT, Sensor Fusion

Hardware & Tools: Oscilloscope, Logic Analyzer, Git, GitHub, Google Test, Fusion 360

Programming : Python, Java, SQL, PHP, Verilo, JavaScript

Web & APIs: REST APIs, MySQL, HTML, CSS

ENGINEERING EXPERIENCE

Robotics Lab Student Lead Robotics Lab, UC Riverside September 2024 – Present

- Architect and deploy embedded robotics systems across **5+ lab projects**, integrating Arduino, Raspberry Pi, motor controllers, multi-sensor arrays, and serial communication; troubleshoot hardware-software integration issues with faculty and students.
- Designed and implemented an equipment access and tracking system integrating magnetic-stripe hardware, SQL databases, and the Google Sheets API, improving equipment accountability and resource utilization across **50+ lab users**.
- Developed and delivered a **6-week Robotics Camp for 20+ students**, teaching embedded C, Python control systems, OpenCV, CNNs, and robotics fundamentals; guided students through building a Bluetooth-controlled robotic car.

Data Science Fellow DZMaC Lab, National Science Foundation, UC Riverside June 2024 – July 2024

- Fine-tuned a DistilBERT LLM classifier to categorize neuroscience transcripts with **83% accuracy**.
- Automated audio preprocessing (noise reduction, segmentation), **cutting transcription time by 72%**.
- Collaborated with neuroscience researchers to integrate model outputs into cognitive recall studies.

INTERN EXPERIENCE

Full Stack Development Intern The Three Logistics, Riverside, CA February 2024 – May 2024

- Developed backend features using REST APIs, enabling address verification, suggestions, and autocomplete functionality
- **Improved shipment processing speed by 25%** through optimized API- based address validation workflows
- Implemented package tracking and account linking using a MySQL database, supporting secure data storage and retrieval
- Built responsive user interfaces and dashboards using JavaScript, PHP, HTML, and CSS

PROJECTS

Autonomous Rover Navigation System | [Github](#) **ESP32-S3 + NavQPlus + ATmega328P** | Oct 2025 – February 2026

- Developed a ROS2 multi-node autonomous navigation system integrating sensing, perception, path planning, and low-level motion control across an ESP32-S3, NavQPlus, and ATmega328P.
- Developed bare-metal C motor drivers on Arduino and implemented PWM-based motor control
- Developed an ESP32-based sensor hub integrating ultrasonic, GPS, IMU, and IR sensors, transmitting real-time data to NavQPlus via CAN bus.
- Implemented GIS path planning and sensor fusion to generate navigation routes across mapped campus environments.
- Integrated NPU-accelerated image segmentation to improve obstacle awareness and environmental perception.

Oura Ring Tamagotchi, [Github](#)

Raspberry Pi Pico + FPGA | May 2026 – June 2026

- Engineered a wearable-integrated embedded system that synchronizes Oura Ring API sleep, activity, and readiness data with a Tamagotchi-style game running on a Raspberry Pi Pico.
- Designed a high-speed SPI interface between a Raspberry Pi Pico and FPGA driving a 480×270 TFT display, with graphics rendered using LVGL.

- Developed FPGA display logic and embedded firmware in Verilog and C to support real-time animations and health-data visualization.

Space Invaders AVR Game | [Github](#)

ATmega328P | May 2025 – June 2025

- Built a real-time game in embedded C on an ATmega1284 using a concurrent task scheduler for game logic, input handling, and display updates.
- Programmed an SPI-driven ST7735 LCD, joystick ADC sampling, and PWM-driven audio output.
- Reduced LCD tearing by optimizing SPI bandwidth and implementing partial screen redraw logic.

Secret Journal, Citrus Hacks [Github](#)

Python + ATmega328P + DeepFace | April 18-19, 2026

- Developed a secure smart journal that authenticates users using facial and voice recognition, building a Python backend with DeepFace and Resemblyzer integrated into a web interface using HTML, CSS, and JavaScript.
- Programmed an Arduino Nano to control a stepper motor locking mechanism, status LEDs, and a 16×2 LCD, establishing UART communication between the embedded hardware and the Python backend.

Computer Vision Motion Detection , [Github](#)

Python + YOLO + OpenCV | February 2026 – March 2026

- Developed a real-time computer vision system to detect, classify, and track pedestrians, cyclists, and scooters from live video streams using a custom-trained YOLO model.
- Built a TensorFlow training pipeline with data augmentation, validation, and custom dataset fine-tuning to improve object detection accuracy across multiple mobility classes.
- Combined classical computer vision techniques—including frame differencing, grayscale thresholding, morphological filtering, contour extraction, and bounding-box overlap analysis—with YOLO detections to identify moving objects and reduce false positives in real time.